8.2 Activities: ratio and scale

← Unit overview · Practice · Answers · Games

1. Lemonade lab

You need: a measuring jug, syrup (*siroop*), water, a few glasses.

Read the bottle: most syrups say something like "1 deel siroop op 5 delen water". Make one glass exactly to the label using a small cup as "1 part". Then ask: "If we want a whole jug, how much syrup and how much water?" Fill in a ratio table together and then make the jug to check.

Try a "too strong" and a "too weak" glass on purpose (1 : 2 and 1 : 10). Taste-test.

Talk about it: "If you double the syrup and double the water, does it taste the same? Why?"

2. Pancake recipe for the whole family

You need: a pancake recipe (or any recipe he likes), paper.

Most Dutch recipes are written *voor 4 personen*. Ask him to rewrite the ingredient list for 2, for 6 and for the number of people actually eating. Encourage the halfway trick: first for 2 (halve everything), then add columns. Then cook it.

Talk about it: "Which ingredient was awkward to change? What did you do with half an egg?"

3. Map of your own street

You need: a printed map of your neighbourhood (any online map can show a scale bar), a ruler, string.

Find home and school. Measure the straight-line distance in cm with a ruler. Use the scale bar or the scale (for example 1 : 25 000, so 1 cm = 250 m) to work out the real distance. Then use string along the actual roads for the "real" route. Compare with what a route planner says.

Extension: make a simple floor plan (*plattegrond*) of his bedroom at 1 : 50 (1 cm = 50 cm). Measure the room with a tape measure first.

Talk about it: "Why is the string route longer than the ruler line?" "Why do maps need a scale at all?"

4. Bike ride speeds

You need: a bike ride you do anyway, a phone or watch to note the time (not to race, just to read off).

Note the distance (from a route planner) and how long the ride took. Afterwards, use a ratio table to work out the average speed in km/u. For example, 4 km in 15 minutes → ×4 → 16 km in 60 minutes → 16 km/u.

Compare with other speeds he knows: walking about 5 km/u, a scooter (*brommer*) 45 km/u, motorway 100 km/u.

Talk about it: "Did you go the same speed the whole way? What does 'average' mean here?"

5. Supermarket best buy

You need: a trip to the supermarket, or a supermarket website, a calculator (optional).

Most Dutch shelf labels show the price per kilo or per litre in small print. Pick two sizes of the same product (peanut butter, hagelslag, rice). He works out per 100 g or per kg first, then checks against the small print on the label.

Talk about it: “Is the biggest pack always the best buy? When might the small pack be better even if it costs more per kilo?” (If you’d throw half away.)

6. Shadow measuring (puzzle activity)

You need: a sunny day, a tape measure, a stick or ruler.

Stand a 1-metre stick upright and measure its shadow. Measure his own shadow, and the shadow of a lamppost or tree. Because all the shadows are made by the same sun at the same moment, the ratio is the same: if the 1 m stick has a 1,5 m shadow and the tree’s shadow is 12 m, the tree is about 8 m tall.

Talk about it: “Why does this only work if we measure at the same moment?”

Conversation prompts for any day

- “The car shows 80 km/u. How far do we go in a quarter of an hour?”
- “This map says 1 : 50 000. How far is 1 cm in real life?”
- “If 3 apples cost €1,50, what does 1 cost? What do 10 cost?”
- “Which is stronger lemonade: 1 part syrup to 4 water, or 2 parts syrup to 10 water?”